1

(a) Nominal GDP: Price (current) × Quantity

In 2017: $10 × 100 + $200 × 50

= 11000

In 2018: $10 × 200 + $200 × 10

= 4000

In 2019: $20 × 200 + $400 × 10

= 8000

Real GDP: Price (base) × Quantity

In 2017: $10 × 100 + $200 × 50

= 11000

In 2018: $10 × 200 + $200 × 10

= 4000

In 2019: $10 × 200 + $200 × 10

= 4000

GDP deflator: (Nominal GDP / Real GDP) × 100

In 2017: $\frac{11000}{11000} \times 100 = 100$

In 2018: $\frac{4000}{4000} \times 100 = 100$

In 2019: $\frac{8000}{4000} \times 100 = 200$

(b) Percentage change in nominal GDP:

In 2018: $\frac{4000-11000}{11000} \times 100$

$= \frac{-7000}{11000} \times 100$

$= 63.64\%$

In 2019: $\frac{8000-4000}{4000} \times 100$

$= \frac{4000}{4000} \times 100$

$= 100\%$

Percentage change in real GDP:

In 2018: $\frac{4000-11000}{11000} \times 100$

$= \frac{-7000}{11000} \times 100$

$= 63.64\%$

In 2019: $\frac{4000-4000}{4000} \times 100$

$= 0\%$

Percentage change in GDP deflator:

In 2018: $\frac{100 - 100}{100} \times 100\%$

$= 0\%$

In 2019: $\frac{200 - 100}{100} \times 100\%$

$= 100\%$

2

(a) Cost of the basket of goods:

In 2017 : $(4 \times \$1) + (2 \times \$2)$

$= 4 + 4$

$= 8$

In 2018 : $(4 \times \$2) + (2 \times \$3)$

$= 8 + 6$

$= 14$

In 2019 : $(4 \times \$3) + (2 \times \$4)$

$= 12 + 8$

$= 20$

Consumer price Index:

In 2017 : $\frac{(4 \times 1) + (2 \times 2)}{(4 \times 1) + (2 \times 2)} \times 100 = 100$

In 2018 : $\frac{(4 \times 2) + (2 \times 3)}{(4 \times 1) + (2 \times 2)} \times 100 = 175$

In 2019 : $\frac{(4 \times 3) + (2 \times 4)}{(4 \times 1) + (2 \times 2)} \times 100 = 250$

(b) Inflation rate:

In 2018 : $\frac{175-100}{100} \times 100 = 75\%$

In 2019 : $\frac{250-175}{175} \times 100 = 42.9\%$

3

Labor force = No. of employed + No. of unemployed

= (125.8 + 10.9) million

= 136.7 million

Unemployment rate = $\frac{\text{No. unemployed}}{\text{Labor force}} \times 100$

= $\frac{10.9}{136.7} \times 100$

= 7.97 %

Adult population = Labor force + Not in labor force

= 136.7 + 79.2

= 215.9 million

Labor force participation rate = $\frac{\text{Labor force}}{\text{population}} \times 100\%$

= (136.7/215.9) × 100 %

= 63.3%

4

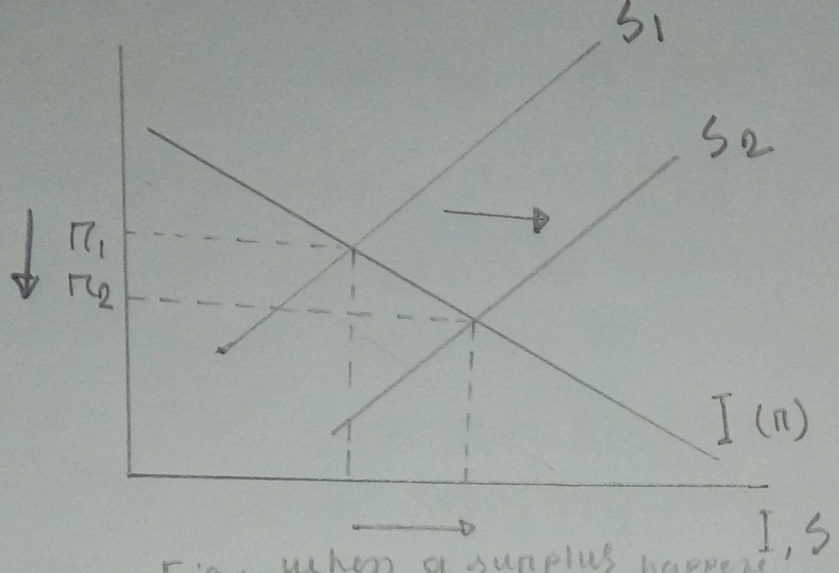


Fig: when a surplus happens in a balanced budget

A budget surplus is having more income than expenses during certain period of time. As the government increases tax, the government borrows less loan from bank. This situation increases the supply of loan able funds, reduces the interest rate and simultaniously increases investment.

Higher investment means greater capital and economic growth.

Budget surplus:

Tax ↑ Govt. takes loan ↓ Interest rate ↓ Investment ↑ GDP ↑

Monetary policy is the process by which the monetary authority manages the money supply to achieve certain goals. There are some tools of monetary policy that can be used for an expansionary monetary policy in a country. They are discount rate, required reserve ratio, Open market operation.

Discount rate: The interest rate the central bank charges on the commercial banks. If the central bank reduces the cost of borrowing, subsequently the banks lower the interest rate they charge for loan. Therefore, the money supply increases.

Required reserve ratio: The percentage of deposit that is required commercial bank to keep as reserve in the central bank. In order to increase money supply the central bank reduces reserve requirements. Commercial bank would see extra funds to be lent out to their client. The banks lower the interest rate and it stimulate consumption and investment expenditure.

Open market operation: The purchase of government securities by the central bank increases money supply. The more money it spends on market the money supply increases.